Hankel-Structured Gerchberg–Saxton Channel Estimation for Rydberg Atomic MIMO Receivers: Bounds, Aperture Scaling, Effective-Rank Normalisation, and Prospective Cross-Generator Validation

(Master version — complete research record for review)

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***Abstract*—Rydberg atomic receivers read out radio fields through an optical transition whose response is proportional to field magnitude, so multi-user channel estimation becomes a phase-retrieval problem with a known additive reference. We first show that, under the unconstrained row-wise parameterisation of the channel, the exact Rician Fisher information is block diagonal across receive elements: each magnitude measurement contributes rank-one information, and the per-element estimation problem is unchanged by adding elements. Increasing the aperture therefore does not by itself increase normalised per-element information for an unstructured estimator, and exploiting aperture requires cross-element structure. On a uniform linear array the geometric multipath model supplies exactly such structure: each path contributes one spatial complex exponential, so the Hankel lifting of every channel column has rank at most the path count. HS-GS enforces this by interleaving a Cadzow structured low-rank projection with an exact expectation–maximisation Gerchberg–Saxton iteration, selecting the model order from held-out pilots rather than an oracle. Against a rank-one Rician Cramér–Rao bound and a geometry-constrained bound on the $3\sum_k L_k$-parameter manifold, HS-GS improves on EM-GS by 2.27–3.04 dB at $N = 32$, and its advantage grows with aperture (-0.19, $+0.78$, $+2.85$ dB at $N = 8, 16, 32$) while EM-GS moves by 0.012 dB. A controlled path-count sweep establishes the mechanism; replacing raw path count by the effective Hankel rank relative to capacity generalises it, giving a sign boundary near $r_{\mathrm{eff}}/r_{\max} \approx 0.5$ that is stable to within 0.07 across $N \in \{16, 32, 64\}$ although the gain magnitude does not collapse. Used prospectively and without refitting, that relation predicted the gain on two previously unused channel generators. Negative results are retained throughout: $N = 8$ is a mixed result, the projection is approximate and non-idempotent, the magnitude collapse is imperfect, a pre-registered K-invariance falsifier is missed by 0.006 dB, and the second prospective prediction errs by 0.315 dB. All results are simulation only, and neither bound strictly governs the biased estimators compared against it.**

***Index Terms*—Rydberg atomic receiver, channel estimation, phase retrieval, Hankel matrix, structured low-rank approximation, constrained Cramér–Rao bound, effective rank.**

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This is a long-form master version assembled for advisor review. It deliberately retains all defensible theory, methodology, diagnostics, positive results, negative results and limitations, and is not compressed to a conference page limit.

I. Introduction

RYDBERG atomic receivers detect radio-frequency fields by driving alkali atoms into highly excited states and reading the resulting optical transmission. The measured quantity is proportional to the *magnitude* of the total incident field superposed on a known local reference beam; the phase is not observed [1], [2], [3]. For a multi-user uplink this turns channel estimation from a linear-Gaussian problem into a phase-retrieval problem with a known additive offset.

Two families of solver exist for this. Alternating-projection methods of the Gerchberg–Saxton type [4] iterate between a measurement constraint and a model constraint; the expectation–maximisation refinement (EM-GS) replaces the hard magnitude substitution by the Rician posterior mean and is the strongest classical baseline we are aware of for this observation model. Separately, recent work unrolls such an iteration into a trained network [5], and a rank projection has been placed inside projected gradient descent for the same problem [6].

The gap. All of these treat the channel matrix $\mathbf{G}$ as an unconstrained collection of parameters. That has a consequence which is easy to miss and, once seen, organises the rest of this paper: under the unconstrained row-wise parameterisation, row n of $\mathbf{GS}$ enters only measurement row n, so the Fisher information is block diagonal across receive elements (§VI). The per-element estimation problem is therefore unchanged by adding elements, and *a larger aperture does not by itself buy an unstructured estimator anything under a normalised metric*. We measure exactly this: EM-GS moves by 0.012 dB across a fourfold change in N (§IX).

Exploiting aperture consequently requires *cross-element* structure. The geometric multipath model on a uniform linear array (ULA) supplies one: each propagation path contributes a single spatial complex exponential across the array, so the Hankel lifting of each channel column is a sum of L_k rank-one terms and has rank at most L_k (§IV). Enforcing that rank inside the iteration is the method studied here, HS-GS. We emphasise at the outset that Hankel structure is *one* such cross-element structure and we make no claim that it is the only one.

Why effective rank becomes necessary. A controlled experiment sweeping the true path count L establishes the mechanism cleanly (§X). But L is a parameter of one generator, not a property of a channel, and it is a poor predictor across generators: a 16-path channel on a 32-element array is algebraically full rank yet has an effective Hankel rank of 8.73, so its lifting is compressible with nothing sparse about it. Replacing L by the Roy–Vetterli effective rank [7] relative to the structural capacity gives a coordinate that can be computed for any channel, and it is what makes the prospective tests of §XIV possible.

A. Contributions

We separate these deliberately, and we do not claim novelty for the ingredients individually — Cadzow projection, EM-GS and constrained Cramér–Rao bounds are all standard.

- **A. Method.** HS-GS: a Cadzow structured low-rank projection interleaved with an exact EM-GS iteration, with the model order chosen from held-out pilots and no oracle path count anywhere. The magnitude observation is never linearised. §V.
- **B. Analysis.** The exact Rician Fisher information for this observation model, showing that each magnitude measurement contributes *rank-one* information rather than rank two, that the resulting matrix is block diagonal across receive elements, and a geometry-constrained bound in tangent-space form because the geometric parameterisation is rank deficient. §VI.
- **C. Mechanism.** Aperture scaling with an unstructured baseline that is flat in N, plus a controlled path-count sweep in which the gain decays monotonically to zero exactly as the prior becomes vacuous — which rules out generic denoising. §IX, §X.
- **D. Generalisation.** Effective Hankel rank relative to capacity as the structural coordinate, and an empirical sign boundary near $r_{\text{eff}}/r_{\max} \approx 0.5$ that is stable across $N \in \{16, 32, 64\}$ — together with the explicit statement that the gain *magnitude* does not collapse onto the same axis. §XII, §XIII.
- **E. Validation.** Two prospective cross-generator predictions, each registered in version control before the corresponding measurement was run, using a stated interpolation rule that was not refitted afterwards. §XIV.
- **F. Robustness and limits.** A K-invariance test at controlled pilot adequacy, the selector's abstention behaviour, and an explicit catalogue of the regimes in which the method fails or the evidence is incomplete. §XV, §XVIII.

II. Related Work

Rydberg atomic receivers and magnitude-only estimation. Cui *et al.* [1] formalise the atomic MIMO observation model in which the receiver measures $|\mathbf{GS}+\mathbf{B}+\mathbf{W}|$, and give the SNR and reference-to-signal-ratio conventions we adopt. Overviews of the technology and its sensing applications appear in [2], [3], and the multi-user uplink is treated in [8]. *These works establish the measurement model; they do not impose channel structure inside the estimator*, which is what this paper does.

Phase retrieval and GS-type solvers. The Gerchberg–Saxton iteration [4] is the classical alternating-projection method for recovering phase from magnitude. Its EM refinement replaces the hard magnitude substitution by the Rician conditional mean and is our baseline. *Existing GS/EM-GS treatments estimate an unconstrained signal*; we retain the exact iteration and add a structural projection.

Unrolled and rank-projected estimators. Xiao *et al.* [5] unroll a stabilised EM-GS into a trained network for this exact receiver. Xu *et al.* [6] place a rank projection inside projected gradient descent and report that "when the pilot length is small ($P = 10$), the PGD performs better than the basic GD, as the projection operation in PGD helps to exploit the rank constraint". *The distinction that matters is which matrix is constrained*: their constraint applies to the channel matrix of a two-dimensional (8×8) array, limiting the number of distinct spatial signatures across users; ours applies per user, to the Hankel lifting of a single column, and encodes the harmonic structure of the ULA manifold, so it is informative even at $K = 1$. The two are complementary. They are also not available at the same scale: their one-dimensional array has $I = 8$ elements, for which $r_{\max}(8) = 4$ by (8), so the embedding admits at most four ranks and the regime this paper characterises barely exists there.

Structured low-rank approximation. Cadzow's composite property mapping [9] alternates projection onto fixed-rank matrices and onto Hankel matrices; Markovsky [10] surveys the structured low-rank approximation problem and its non-convexity. *We use these as given* and do not claim a new projection algorithm; our contribution is placing the projection inside the exact phase-retrieval iteration and selecting its order without an oracle.

Constrained Cramér–Rao bounds. Gorman and Hero [11] and Stoica and Ng [12] give the constrained CRLB in the form we require, which depends on a basis of the tangent space rather than on the parameterisation itself — necessary here because the geometric parameterisation is rank deficient (§VI-B).

Effective rank. Roy and Vetterli [7] define the effective rank as the exponential of the spectral entropy of the normalised singular values. *We use it as a diagnostic coordinate*, not as a new theoretical object.

Channel model. The clustered Saleh–Valenzuela model used in the cross-generator tests follows the description in [13], which is the reference [5] itself cites; the second configuration is taken from the simulation section of [14], a precoding study by the same group as [1].

III. System Model

A. Observation model

An N-element ULA serves K single-antenna users over P pilot symbols. Let $\mathbf{G} \in \mathbb{C}^{N\times K}$ be the channel, $\mathbf{S} \in \mathbb{C}^{K\times P}$ the pilot matrix, $\mathbf{B} \in \mathbb{C}^{N\times P}$ the known reference beam and $\mathbf{W} \in \mathbb{C}^{N\times P}$ additive noise. The receiver observes

$$\mathbf{Z} = \big|\mathbf{GS} + \mathbf{B} + \mathbf{W}\big|, \qquad \mathbf{W}_{np} \sim \mathcal{CN}(0, \sigma^2), \tag{1}$$

with $|\cdot|$ applied elementwise and $\mathbf{Z} \in \mathbb{R}_{\geq 0}^{N\times P}$.

Two properties of (1) are load-bearing. First, **the noise is inside the modulus**: $\mathbf{Z}$ is not a noisy observation of $|\mathbf{GS}+\mathbf{B}|$, and each entry is Rice distributed rather than Gaussian. Second, **the modulus is never linearised** anywhere in HS-GS or in either baseline; this is checked per trial by the verification harness (§VII).

A terminological clarification. Throughout, "biased phase retrieval" refers to the known reference-field offset $\mathbf{B}$ in (1). This is entirely distinct from the *statistical estimator bias* discussed in §VI-C, which is why the Cramér–Rao bounds do not strictly govern the estimators compared against them.

B. Geometric ULA channel

User k's channel follows a sparse geometric multipath model with L_k paths:

$$g_k[n] = \sum_{\ell=1}^{L_k} \alpha_{\ell,k}\, e^{-\jmath(n-1)\psi_{\ell,k}}, \qquad n = 1,\ldots,N, \tag{2}$$

with, for half-wavelength element spacing,

$$\psi_{\ell,k} = \pi \sin\theta_{\ell,k}. \tag{3}$$

Equivalently $\mathbf{g}_k = \sum_\ell \alpha_{\ell,k}\mathbf{a}(\theta_{\ell,k})$ with $[\mathbf{a}(\theta)]_n = e^{-\jmath\pi(n-1)\sin\theta}$.

The assumptions, stated so they can be attacked:

- *Sparse geometric multipath.* L_k is small relative to N; in the principal experiments $L_k \sim \mathcal{U}\{3,\ldots,7\}$ [6].
- *Path gains.* $\alpha_{\ell,k}$ are i.i.d. circularly symmetric complex Gaussian with equal expected power per path.
- *Angles of arrival.* $\theta_{\ell,k}$ i.i.d. uniform on $(-90°, 90°)$ in the principal generator. The cross-generator tests of §XIV change this distribution and only this kind of thing.
- *Narrowband, flat.* No delay spread enters (1).
- *Equal user power.* All users transmit at the same power; near–far effects are untested (§XVIII).
- *Pilots.* $\mathbf{S}$ is drawn once per trial and shared bit-identically by every estimator being compared (§VII).

C. SNR and RSR conventions

SNR is defined at the array input in the usual sense. The reference-to-signal ratio (RSR) is defined against a *single* user, as in [1]. Since K users transmit simultaneously, a nominal single-user RSR of 12 dB corresponds to $12 - 10\log_{10} K = 7.23$ dB against the K-user total at $K = 3$. The empirically measured conversion is slightly larger than the nominal 4.77 dB — 4.85 dB over 300 realisations — which places the same case at 7.15 dB when measured rather than computed. The difference is finite-sample; we quote nominal values and all RSR figures below are single-user.

D. Three experiment families, not one

Results in this paper come from three distinct experiment families with different configurations. **They are never pooled**, and every figure and table below names the family it belongs to. Table I gives the model parameters and Table II the per-family configurations.

TABLE I
SYSTEM AND MODEL PARAMETERS.

Symbol	Meaning / value
N	receive elements; $8, 16, 32$ (B3, EXT), $16, 32, 64$ (TD)
K	users; 3 unless swept $(2, 3, 4$ in §XV)
P	pilot symbols; see Table II
$\mathbf{G}$	$N \times K$ channel, columns as in (2)
$\mathbf{S}$	$K \times P$ pilots, shared across compared estimators
$\mathbf{B}$	$N \times P$ known reference beam
$\mathbf{W}$	$N \times P$ noise, $\mathcal{CN}(0, \sigma^2)$, inside the modulus
L_k	paths per user, $\mathcal{U}\{3,\ldots,7\}$ [6]
$\theta_{\ell,k}$	AoA, $\mathcal{U}(-90°, 90°)$
$r_{\max}(N)$	Hankel capacity $= \lceil N/2 \rceil$, (8)

IV. Why Hankel Structure Exists

Drop the user index and write one channel column as $g_n = \sum_{\ell=1}^{L} \alpha_\ell z_\ell^n$ with $z_\ell = e^{-\jmath\psi_\ell}$. Define the Hankel lifting with pencil parameter p,

$$\mathcal{H}(\mathbf{g}) = \begin{bmatrix} g_0 & g_1 & \cdots & g_p \\ g_1 & g_2 & \cdots & g_{p+1} \\ \vdots & & \ddots & \vdots \\ g_{N-p-1} & \cdots & & g_{N-1} \end{bmatrix} \in \mathbb{C}^{(N-p)\times(p+1)}. \tag{4}$$

Its (q, r) entry is g_{q+r}, which separates in q and r:

$$\big[\mathcal{H}(\mathbf{g})\big]_{q,r} = \sum_{\ell=1}^{L} \alpha_\ell z_\ell^{q+r} = \sum_{\ell=1}^{L} \alpha_\ell\, z_\ell^{q}\, z_\ell^{r}. \tag{5}$$

With $\mathbf{v}_\ell = [1, z_\ell, \ldots, z_\ell^{N-p-1}]^T$ and $\mathbf{w}_\ell = [1, z_\ell, \ldots, z_\ell^{p}]^T$ this is

$$\mathcal{H}(\mathbf{g}) = \sum_{\ell=1}^{L} \alpha_\ell\, \mathbf{v}_\ell \mathbf{w}_\ell^T, \tag{6}$$

a sum of L rank-one terms, hence

$$\operatorname{rank}\big(\mathcal{H}(\mathbf{g})\big) \leq L. \tag{7}$$

Each propagation path creates one spatial complex exponential across the ULA and therefore contributes exactly one rank-one component to the lifting. Three consequences deserve emphasis.

When equality holds. If the z_ℓ are distinct and L does not exceed either matrix dimension, (7) holds with equality; conversely a sequence whose Hankel matrix has rank L is a sum of L exponentials [9].

Closely spaced paths. If two z_ℓ are nearly equal the corresponding rank-one terms are nearly collinear, so the matrix is only *numerically* rank deficient: the spectrum decays rather than terminating in an exact cliff. This is precisely the regime the clustered generator of §XIV creates, and it is why raw path count eventually fails as a predictor (§XII).

Grid-free, and no angle estimation. The constraint is on the rank of (4), not on coefficients over a discretised angular dictionary. No $\theta_{\ell,k}$ is ever estimated, and no basis mismatch is possible.

TABLE II
THE THREE EXPERIMENT FAMILIES. THESE ARE SEPARATE EXPERIMENTS WITH DIFFERENT CONFIGURATIONS AND ARE NEVER POOLED.

Family	Store / driver	Configuration	Statistic	Used in
B3 (principal)	`results/track_b/b3/` `scripts/plot_paper.py`	$N \in \{8, 16, 32\}$, $P \in \{10, 30\}$, SNR $\{-5, 0, 5, 10, 15, 20\}$, RSR 12 dB, $T = 50$, $n_{\mathrm{cz}} = 4$; 400 trials/point	ratio-of-sums; per-point mean across points	Figs. 1, 2; §V-E, VIII, IX
EXT (diagnostic)	`trackB_hankel_emgs/` `figure_array_size_` `comparison.py`	$P = 30$ only, SNR $\{-10, \ldots, 20\}$ (7 points), RSR 12 dB, $T = 50$, $n_{\mathrm{cz}} = 4$; 600/400/200 trials	ratio-of-sums; mean of per-SNR dB values	Figs. 3, 4, 5, 6; §IX-C, X, XI
TD (effective rank)	`results/track_d/partB9/` `scratch/trackD_partB9_sweeps.py`	$N \in \{16, 32, 64\}$, $P = 20$, SNR $\sim \mathcal{U}[-10, 20]$, RSR 10 dB, $T = 100$, $n_{\mathrm{cz}} = 1$, adaptive order	paired per-trial median, bootstrap CI	Figs. 7, 8, 9; §XIII–XV

A. Structural capacity

Since $\mathcal{H}(\mathbf{g})$ is $(N-p) \times (p+1)$, its rank cannot exceed $\min(N-p, p+1)$, maximised over p at

$$r_{\max}(N) = \max_{1 \le p \le N-1} \min(N-p,\, p+1) = \lceil N/2 \rceil, \qquad (8)$$

attained at $p = \lfloor N/2 \rfloor$. Thus

$$r_{\max}(8) = 4, \quad r_{\max}(16) = 8, \quad r_{\max}(32) = 16, \quad r_{\max}(64) = 32.$$

The ceiling, not the floor: at $N = 7$ the maximum is 4, and the two expressions coincide only for even N.

If $\hat{L} \ge r_{\max}$ then $\operatorname{rank} \le \hat{L}$ holds for *every* length-N vector, so the constraint carries no information whatever and the projection is the identity map. The capacity therefore states precisely *when* the prior can be informative, and it is the natural denominator for any rank-like quantity — which is the role it plays in §XII.

B. Degrees-of-freedom motivation, and what it does not prove

Treated as unstructured, $\mathbf{G}$ has $2NK$ real degrees of freedom. Under (2) it is described by approximately $3\sum_k L_k$ real parameters (two per complex gain, one per angle). With $L_k \equiv \bar{L}$ the compression ratio is

$$\frac{3\sum_k L_k}{2NK} = \frac{3K\bar{L}}{2NK} = \frac{3\bar{L}}{2N}, \qquad (9)$$

in which K cancels.

What this motivates and what it does not. It motivates the hypothesis that the structural gain should be invariant in K once pilot adequacy is held fixed, because the projection acts per user column and the compression available per column does not depend on how many columns there are. It does *not* prove that hypothesis: a parameter count says nothing about estimation error, the bound geometry, or the behaviour of a particular iterative algorithm. §XV tests it directly, and the test partially fails.

V. THE HS-GS ESTIMATOR

A. Exact EM-GS measurement update

Let $\mathbf{\Lambda} = \mathbf{GS} + \mathbf{B}$ be the current noiseless-field estimate. Conditional on it, each observation z is Rice distributed with non-centrality $a = |\lambda|$, and the conditional mean of the complex field is obtained by scaling the observed magnitude by the Bessel ratio $R(x) = I_1(x)/I_0(x)$ evaluated at $\kappa = 2za/\sigma^2$. Algorithm 1 states one iteration. The row-wise least squares in line 4 is exactly the structure that makes the Fisher information block diagonal (§VI-A).

Algorithm 1 EM-GS measurement update $\mathcal{T}_{\mathrm{EM}}$ (one iteration)

Require: $\mathbf{Z}, \mathbf{S}, \mathbf{B}, \sigma^2$; iterate $\mathbf{G}$ ($\varnothing \Rightarrow$ spectral init)
1: $\mathbf{\Lambda} \leftarrow \mathbf{GS} + \mathbf{B}$; $\quad \boldsymbol{\kappa} \leftarrow (2/\sigma^2)\, \mathbf{Z} \odot |\mathbf{\Lambda}|$
2: $\mathbf{R} \leftarrow R(\boldsymbol{\kappa})$, $R(x) = I_1(x)/I_0(x)$ ▷ elementwise
3: $\hat{\mathbf{Y}} \leftarrow \mathbf{R} \odot \mathbf{Z} \odot e^{\jmath \angle \mathbf{\Lambda}}$ ▷ Rician posterior mean
4: $\mathbf{G}_{n,:} \leftarrow (\hat{\mathbf{Y}}_{n,:} - \mathbf{B}_{n,:})\mathbf{S}^H(\mathbf{SS}^H)^{-1}$, $n = 1..N$ ▷ rows independent
5: **return** $\mathbf{G}$

B. Cadzow projection

Let $\mathcal{S}_{\hat{L}}$ be the set of length-N sequences whose Hankel lifting has rank at most $\hat{L}$. Membership is the intersection of two sets that each admit an exact projector: fixed-rank matrices (truncated SVD, Frobenius-optimal by Eckart–Young) and Hankel matrices (anti-diagonal averaging). Cadzow's algorithm [9] alternates them (Algorithm 2).

The projection is approximate, and we do not hide this. The intersection is a non-convex variety. The composite map

$$\mathcal{P}_{\hat{L}} = \mathcal{H}^{\dagger} \circ \Pi_{\hat{L}} \circ \mathcal{H} \qquad (10)$$

is *not* in general an idempotent projection onto that intersection: one application lands near, but not on, the set, and iterating changes the result. This is why $n_{\mathrm{cz}} = 4$ sweeps are used rather than one — more sweeps move closer to the intersection — and why no convergence guarantee is claimed for the outer iteration. §XI quantifies the residual directly: after four sweeps the singular-value cliff is restored only to $\sim 10^{-3}$, against $\sim 10^{-16}$ for the true channel.

When $\hat{L} \ge r_{\max}$ the routine returns its input unchanged, so HS-GS then reproduces EM-GS bit for bit (Algorithm 2, line 2).

C. Interleaved iteration

The projection acts *inside* the exact iteration, not as post-processing:

$$\mathbf{G}^{(t)} = \mathcal{P}_{\hat{L}} \circ \mathcal{T}_{\mathrm{EM}}\big(\mathbf{G}^{(t-1)}\big), \qquad t = 1, \ldots, T, \qquad (11)$$

with $\mathcal{P}_{\hat{L}}$ applied column-wise after every one of the T updates.

Why inside rather than after. The EM-GS update quality depends on the current iterate through $\kappa = 2z|\lambda|/\sigma^2$: the Bessel ratio is evaluated at the current field estimate, so a structurally cleaner iterate produces a better phase estimate, hence a better next iterate. Post-processing a converged unstructured estimate cannot recover this compounding, because the phase estimates that produced it were never improved.

Algorithm 2 Cadzow projection $\mathcal{P}_{\hat{L}}$ of one channel column

Require: $\mathbf{g} \in \mathbb{C}^N$, rank $\hat{L}$, sweeps n_{cz}

$p \leftarrow \arg\max_p \min(N-p, p+1)$; $\quad r_{\max} \leftarrow \lceil N/2 \rceil$
if $\hat{L} \geq r_{\max}$ **then return g** $\quad \triangleright$ constraint vacuous: exact no-op
end if
for $t = 1..n_{\mathrm{cz}}$ **do**
$\quad \mathbf{H} \leftarrow \mathcal{H}(\mathbf{g})$, $\mathbf{H}_{q,r} = g_{q+r}$ $\quad \triangleright$ $(N-p) \times (p+1)$
$\quad \mathbf{U}\boldsymbol{\Sigma}\mathbf{V}^H \leftarrow \mathrm{svd}(\mathbf{H})$; $\ \mathbf{H} \leftarrow \mathbf{U}_{:,1:\hat{L}} \boldsymbol{\Sigma}_{1:\hat{L}} \mathbf{V}^H_{:,1:\hat{L}}$ $\quad \triangleright$ rank
$\quad g_m \leftarrow |\mathcal{A}_m|^{-1} \sum_{(q,r)\in\mathcal{A}_m} \mathbf{H}_{q,r}$, $\mathcal{A}_m = \{q+r=m\}$ $\quad \triangleright$ Hankel
end for
return g

Algorithm 3 HS-GS, and the held-out rank selection it uses

1: **function** HS-GS($\mathbf{Z}, \mathbf{S}, \mathbf{B}, \sigma^2$; $\hat{L}$, T)
2: $\quad$ active $\leftarrow (\hat{L} < r_{\max})$; $\quad \mathbf{G} \leftarrow \varnothing$
3: $\quad$ **for** $t = 1..T$ **do**
4: $\quad\quad \mathbf{G} \leftarrow \mathcal{T}_{\mathrm{EM}}(\mathbf{G})$ $\ \triangleright$ Alg. 1: *identical* to the baseline
5: $\quad\quad$ **if** active **and** $t \bmod \Pi = 0$ **then**
6: $\quad\quad\quad \mathbf{G}_{:,k} \leftarrow \mathcal{P}_{\hat{L}}(\mathbf{G}_{:,k})$, $k = 1..K$ $\quad \triangleright$ Alg. 2
7: $\quad\quad$ **end if**
8: $\quad$ **end for**
9: $\quad$ **return G**
10: **end function**

11: **function** SELECTRANK($\mathbf{Z}, \mathbf{S}, \mathbf{B}, \sigma^2$; ν, T_{sel})
12: $\quad \mathcal{V} \leftarrow$ last $\lceil \nu P \rceil$ pilot columns; $\quad \mathcal{F} \leftarrow$ the rest
13: $\quad$ **for** $L = 1..r_{\max}$ **do**
14: $\quad\quad \mathbf{G}_L \leftarrow$ HS-GS$(\mathbf{Z}_{:,\mathcal{F}}, \mathbf{S}_{:,\mathcal{F}}, \mathbf{B}_{:,\mathcal{F}}, \sigma^2; L, T_{\mathrm{sel}})$
15: $\quad\quad J(L) \leftarrow \|\mathbf{Z}_{:,\mathcal{V}} - |\mathbf{G}_L \mathbf{S}_{:,\mathcal{V}} + \mathbf{B}_{:,\mathcal{V}}|\|_F^2$ $\quad \triangleright$ columns never fitted
16: $\quad$ **end for**
17: $\quad$ **return** $\hat{L} = \arg\min_L J(L)$
18: **end function**

D. Held-out order selection

HS-GS needs a target rank $\hat{L}$, and the true L_k is unknown to any practical receiver.

The in-sample residual cannot supply it. Constraining $\mathbf{G}$ can only worsen the fit to the pilots used for fitting, so an in-sample criterion always prefers the largest candidate and the selector would degenerate to $\hat{L} = r_{\max}$, i.e. to EM-GS. Holding out a validation fraction $\nu = 0.3$ of the pilot columns penalises over-modelling correctly, and every quantity involved is observable:

$$\hat{L} = \arg\min_{1 \leq L \leq r_{\max}} \left\| \mathbf{Z}_{\mathcal{V}} - \left| \mathbf{G}_L \mathbf{S}_{\mathcal{V}} + \mathbf{B}_{\mathcal{V}} \right| \right\|_F^2, \tag{12}$$

with $\mathbf{G}_L$ the estimate at order L from the fitting columns only, run for $T_{\mathrm{sel}} = 20$ iterations. **No oracle rank is used anywhere in this paper.**

The selector returns one scalar shared by all K users, which is a limitation we state rather than defend (§XVIII). Order matters both ways: $\hat{L} < L$ truncates genuine components, a bias that does *not* vanish as noise decreases, while $\hat{L} > L$ admits noise and weakens the prior. Ours systematically under-selects: true orders $(7, 4, 6)$ typically give $\hat{L} = 5$, and the mean $\hat{L}$ reaches only 11.63 at $L = 16$ (§X).

Abstention. When the selector returns $\hat{L} \geq r_{\max}$ the projection is the identity and HS-GS reduces exactly to EM-GS. This is not a failure mode but a designed one: the rule declines to impose structure it cannot support from the held-out residual, without being told which channel it faces. §XIV measures how often it does so on two different generators.

Relation to the baseline. Deleting lines 5–7 of Algorithm 3 yields baseline EM-GS exactly: both chain the same $\mathcal{T}_{\mathrm{EM}}$ for the same T iterations from the same initialisation, with no early stopping in either. Setting $\hat{L} \geq r_{\max}$ makes $\mathcal{P}_{\hat{L}}$ the identity, so the two then agree *bit for bit*; this is checked numerically (§VII).

E. Complexity

One EM-GS update costs $O(NPK)$. The projection adds Kn_{cz} SVDs per iteration, $O(Kn_{\mathrm{cz}}N^3)$, and order selection multiplies this by the $r_{\max}$ candidates.

Measured against a *chained* EM-GS baseline — the correct reference, since HS-GS issues T single-iteration calls — HS-GS at fixed order costs 1.49, 1.32, 1.22× at $N = 8, 16, 32$, so the ratio *falls* with N: the $O(N^3)$ SVD grows more slowly than the accumulated cost of the baseline iteration at these sizes. Order selection dominates total runtime, 57.7–85.5%, which makes the candidate set the obvious practical target — for instance a coarse-to-fine search, or reusing $\hat{L}$ across coherence blocks. We describe this rather than pursue it.

VI. Exact Likelihood and Performance Bounds

A. Rank-one Fisher information

Conditional on $\mathbf{G}, \mathbf{S}, \mathbf{B}$ the noiseless field at (n, p) is a known constant $\lambda = (\mathbf{GS} + \mathbf{B})_{n,p}$, so $z = |\lambda + w|$ is Rice distributed:

$$p(z \mid \lambda) = \frac{2z}{\sigma^2} \exp\Big(-\frac{z^2 + |\lambda|^2}{\sigma^2}\Big) I_0\Big(\frac{2z|\lambda|}{\sigma^2}\Big). \tag{13}$$

Crucially (13) depends on λ *only through* $a = |\lambda|$. Differentiating, with $\frac{d}{dx} \log I_0(x) = R(x)$,

$$\frac{\partial \log p}{\partial a} = \frac{2}{\sigma^2}\big[zR(\kappa) - a\big], \qquad \kappa = \frac{2za}{\sigma^2}. \tag{14}$$

Using the zero-mean-score identity $\mathbb{E}[zR(\kappa)] = a$ and writing $\beta(a, \sigma^2) = \sigma^{-4}\big(\mathbb{E}[z^2R^2(\kappa)] - a^2\big)$, one measurement carries scalar information $I_a = \mathbb{E}[(\partial_a \log p)^2] = 4\beta$. In real coordinates $\boldsymbol{\eta} = [\Re \mathrm{vec}\,\mathbf{G}; \Im \mathrm{vec}\,\mathbf{G}]$ its contribution to the Fisher matrix is

$$\mathbf{J}_n = \sum_{p=1}^{P} 4\beta(a_{n,p}, \sigma^2)\, \mathbf{u}_{n,p}\mathbf{u}_{n,p}^T, \qquad \mathbf{u}_{n,p} = \frac{\partial a_{n,p}}{\partial \boldsymbol{\eta}_n}. \tag{15}$$

Each magnitude measurement contributes rank-one information, not rank two. This follows directly from (13) depending on λ only through its modulus: the measurement constrains one real direction, not two. Crediting both quadratures — as a naive complex-Gaussian analogy would — overstates the information and yields a bound too low by 0.17–4.64 dB across the 36 operating points of family B3.

An identifiability check settles the form. At $K=3$, $P=3$ the per-row problem is unidentifiable — three complex unknowns from three real measurements — and (15) is correctly singular, whereas the rank-two construction claims full rank. A construction that reports an identifiable problem where none exists is wrong regardless of any numerical comparison.

Block diagonality. Because row n of $\mathbf{GS}$ enters only measurement row n (Algorithm 1, line 4), $\mathbf{J}$ is block diagonal across receive elements. Under this unconstrained row-wise parameterisation the per-element estimation problem is therefore unchanged by adding elements, and normalised per-element information does not grow with N.

Scope, stated precisely. This is a statement about the unconstrained row-wise parameterisation of (1). It implies that exploiting aperture requires *cross-element* structure. It does *not* imply that Hankel structure is the only such structure, nor that any particular structured estimator must benefit. HS-GS uses the Hankel structure induced by (2); other cross-element priors could serve.

B. Constrained bound on the geometric manifold

The unconstrained bound governs estimators treating $\mathbf{G}$ as $2NK$ free parameters. HS-GS instead exploits (2), whose parameters $\boldsymbol{\phi} = \{\psi_{\ell,k}, \Re\alpha_{\ell,k}, \Im\alpha_{\ell,k}\} \in \mathbb{R}^{3\sum_k L_k}$ number about 45 at the default configuration, *independent of* N.

With $\mathbf{D} = \partial\boldsymbol{\eta}/\partial\boldsymbol{\phi}$ the textbook form $\mathbf{D}(\mathbf{D}^T\mathbf{J}\mathbf{D})^{-1}\mathbf{D}^T$ is unusable, because $\mathbf{D}$ is rank deficient. At $N=8$: its mean numerical rank is 40.4 against approximately 44.7 parameters, and 28.1% of realisations have $3\sum_k L_k > 2NK$, so $\boldsymbol{\phi} \mapsto \mathbf{G}$ is not injective. We therefore use the Gorman–Hero / Stoica–Ng tangent-space form [11], [12], with $\mathbf{U}$ an orthonormal basis of $\mathrm{range}(\mathbf{D})$ obtained by SVD at tolerance 10^{-9}:

$$\mathrm{CCRB} = \mathbf{U}\big(\mathbf{U}^T\mathbf{J}\mathbf{U}\big)^{-1}\mathbf{U}^T. \tag{16}$$

This depends on the tangent *space* rather than on the particular parameterisation, and is therefore invariant to over-parameterisation — which is exactly the pathology present here. Both bounds average over the same channel realisations as the estimator curves.

C. Bias and the interpretation of the bounds

Both bounds constrain the covariance of *unbiased* estimators, whereas

$$\mathrm{MSE} = \mathrm{variance} + \mathrm{bias}^2. \tag{17}$$

All three estimators here run to a fixed iteration budget with the shrinking Bessel update, and HS-GS additionally truncates rank; none is unbiased, so neither bound applies strictly.

Concretely: the unconstrained CRLB is a reference for GS and EM-GS only, since HS-GS uses prior information the bound does not model; and the CCRB, itself an unbiased-estimation bound, does not forbid a sufficiently biased structured estimator from falling below it. §VIII-B reports both effects.

We therefore do not call EM-GS efficient. What we claim is the weaker and defensible statement that it *numerically tracks* the unconstrained bound closely enough for that bound to be an informative reference.

VII. Experimental Methodology and Reproducibility

A. Paired worlds

Within an operating point, every estimator being compared receives one *read-only* world $(\mathbf{G}, \mathbf{S}, \mathbf{B}, \mathbf{W}, \mathbf{Z})$ — bit-identical channel, pilots, reference beam, noise and observation. Contrasts therefore benefit from cancellation of the common channel draw. Realisations are independent *across* operating points.

B. Two averaging domains, kept distinct

This distinction matters enough to state explicitly, because conflating the two produces numbers that differ by decibels.

Within an operating point, NMSE is pooled as a ratio of sums,

$$\mathrm{NMSE}_G = 10\log_{10}\frac{\sum_t \|\hat{\mathbf{G}}_t - \mathbf{G}_t\|_F^2}{\sum_t \|\mathbf{G}_t\|_F^2}, \tag{18}$$

never a mean of per-trial decibels. Equation (18) estimates the ensemble quantity a Cramér–Rao bound bounds, whereas a mean-of-decibels is smaller by Jensen's inequality. The factor is 10, not 20, the arguments being energies. The paired gain is

$$\Delta_{\mathrm{HS}} = 10\log_{10}\frac{\sum_t \|\hat{\mathbf{G}}_t^{\mathrm{EM}} - \mathbf{G}_t\|_F^2}{\sum_t \|\hat{\mathbf{G}}_t^{\mathrm{HS}} - \mathbf{G}_t\|_F^2}, \tag{19}$$

with a 95% paired bootstrap interval over 2000 resamples.

Across operating points, the mean is taken of the per-point decibel values, *after* conversion. This is necessary because linear NMSE spans three orders of magnitude across the SNR grid, so a linear average across SNR is dominated by the lowest-SNR point (roughly 70% of the sum against 0.72% for the top three) and is nearly blind to the high-SNR half — precisely where the projection hurts at $N=8$.

Rule used throughout: ratio-of-sums wherever a curve is compared to a bound; per-point mean of decibel contrasts when summarising across operating points; paired per-trial median with a bootstrap interval for the Track D contrasts, whose SNR is drawn continuously rather than gridded.

C. Configurations

Table II gives the three families. Within family B3 the rank selector uses $\nu = 0.3$, $T_{\mathrm{sel}} = 20$; the estimator uses $T = 50$ iterations and $n_{\mathrm{cz}} = 4$ Cadzow sweeps with $\Pi = 1$ (projection after every update). Family TD uses adaptive order selection with a single Cadzow sweep and $T = 100$; because one sweep and four sweeps define *different* operators (§V-B), TD and B3 numbers are never compared directly as though they measured the same estimator.

D. Verification

A standalone package reproduces the comparison and passes thirteen automated checks. Two are load-bearing for fairness and are executed rather than asserted:

1) each trial hands both estimators one read-only world; and

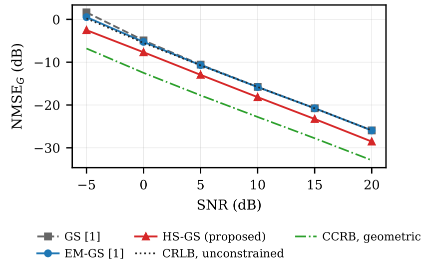

Fig. 1. **Family B3.** NMSE versus SNR at $N = 32$, $K = 3$, $P = 30$, RSR $= 12$ dB (single-user), $T = 50$, $n_{\mathrm{cz}} = 4$, 400 paired trials per point, ratio-of-sums pooling (18). EM-GS tracks the rank-one unconstrained CRLB to within 0.051 dB above 5 dB. Neither bound governs a biased estimator (§VI-C).

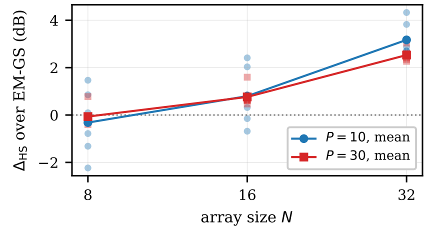

Fig. 2. **Family B3.** Paired HS-GS gain over EM-GS versus array size. $K = 3$, RSR $= 12$ dB, $T = 50$, $n_{\mathrm{cz}} = 4$, 400 trials per point. Two pilot counts are shown: $P = 30$, which Figs. 1 and 4 also use, and $P = 10$ as an auxiliary curve. Faint markers are the six individual SNR operating points of each curve; solid lines are their means. The headline $-0.19/+0.78/+2.85$ dB is the mean over all twelve points (six SNR values $\times$ two pilot counts). At $N = 8$ the points straddle zero, so no single scalar summarises that array size.

2) **with the projection disabled, HS-GS reproduces EM-GS to** $\max|\mathrm{diff}| = 0$ — the projection is provably the only difference between them.

A third confirms §IV: a noiseless L-path column has $\sigma_{L+1}/\sigma_1 \approx 7 \times 10^{-16}$ (check 6: 6.73×10^{-16} at $L = 2$, 7.56×10^{-16} at $L = 3$).

VIII. Baseline Performance and Bounds

A. Main result at $N = 32$, $P = 30$

Fig. 1 shows all three estimators against both bounds at the principal operating configuration.

EM-GS numerically tracks the unconstrained CRLB to within 0.051 dB for SNR ≥ 5 dB. We do not call it efficient — it is biased, and the bound formally governs unbiased estimators (§VI-C) — but the close numerical agreement is what makes the benchmark informative rather than vacuous. At -5 and 0 dB the curves separate by 0.295 and 0.237 dB, consistent with the low-SNR bias of the shrinking update.

The geometric CCRB lies 7.05–7.11 dB below the unconstrained bound at this configuration, quantifying what the structure is worth in principle. HS-GS occupies part of that gap, improving on EM-GS by 2.27–3.04 dB across the sweep.

B. Distance to the bounds, and crossings

HS-GS lies above the geometric CCRB at 34 of the 36 family-B3 points, by 0.34–9.09 dB, and the gap *grows* with N (+0.34–1.44 dB at $N = 8$, $P = 30$; +4.36–4.88 dB at $N = 32$, $P = 30$): HS-GS captures more structural gain in absolute terms as the array grows, but a smaller share of what is available.

Curves do fall below the plotted bounds, in two distinct senses, and neither is a violation.

Below the unconstrained CRLB. HS-GS lies below it at 26 of 36 points, by up to 6.94 dB. That bound governs estimators treating $\mathbf{G}$ as $2NK$ free parameters; HS-GS exploits a prior it does not model, so the bound simply does not apply to it. Its proper reference is the CCRB, respected at 34 of 36.

Below the CCRB. The remaining two crossings are consistent with bias, as (17) permits. They occur at the two lowest-SNR points of the smallest aperture — $N = 8$, $P = 10$, SNR -5 and 0 dB — by 2.35 and 0.15 dB, which is where the shrinking update is furthest from unbiased. The baselines show the same low-SNR pattern and no high-SNR one: GS and EM-GS dip below the CCRB only at $N = 8$, $P = 10$, -5 dB, by 0.55 and 0.88 dB, and stay above it at every point with SNR ≥ 15 dB (nearest margin 0.84 dB).

A per-trial shrinkage diagnostic $c = \langle \hat{\mathbf{G}}, \mathbf{G} \rangle / \|\mathbf{G}\|_F^2$ would make this argument quantitative rather than circumstantial. The retained `b3` archives store squared-error numerators only, so c cannot be recovered from them without rerunning the sweep, and we do not quote it. **We do not claim bias accounts for every crossing** — only that the observed pattern is consistent with it and that no crossing requires an estimator to beat a bound that governs it.

IX. Aperture Scaling

This is the result that connects the Fisher analysis to measurement.

A. Principal result (family B3)

Fig. 2 plots the paired gain against array size. Averaged over the twelve operating points at each array size — the six SNR values of Table II at each of $P \in \{10, 30\}$ — the gain is

$$-0.19, \quad +0.78, \quad +2.85 \text{ dB} \quad \text{at } N = 8, 16, 32,$$

with win rates 33.5, 74.0, 95.3% and the constraint active in 57.5, 92.7, 99.6% of trials.

Under the same averaging EM-GS varies by 0.012 dB. This is the measured counterpart of the block-diagonal Fisher structure of §VI-A: the N-dependence belongs to the structural prior, not to a degrading baseline. Had EM-GS itself deteriorated with N, the widening gap would be uninformative.

B. $N = 8$ is a mixed result, not a small positive one

At $N = 8$ the capacity is $r_{\max} = 4$ while $L_k \in \{3, \ldots, 7\}$, so the prior is frequently vacuous or actively harmful: the projection is inactive in 42.5% of trials, and where active it often truncates genuine components. HS-GS gains +0.78 and +1.47 dB at -5 dB (at $P = 30$ and $P = 10$ respectively) but turns negative as SNR rises, reaching -2.23 dB.

The mechanism is the one (17) predicts. Truncation bias does not shrink with noise. As SNR rises the variance term

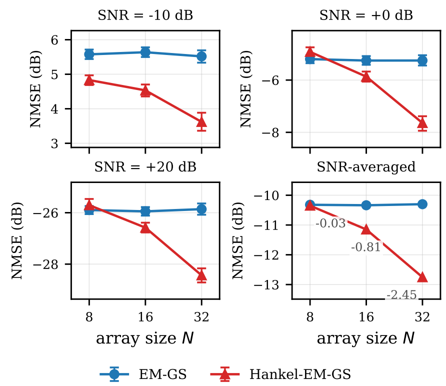

Fig. 3. **Family EXT — a separate experiment from Fig. 2.** Absolute channel NMSE versus array size: three SNR regimes and, bottom right, the SNR-averaged summary (mean of the seven per-SNR decibel values, taken *after* conversion; inset labels give HS-GS minus EM-GS). $K = 3$, $P = 30$ only, RSR $= 12$ dB, $L_k \sim \mathcal{U}\{3..7\}$, SNR grid extended to -10 dB; $600/400/200$ paired trials per point at $N = 8/16/32$; error bars 95% bootstrap.

falls while the bias term does not, so the trade stops paying and then reverses. **The sign of any scalar summary at $N = 8$ depends on the pooling**, which is why we report the per-point values rather than a single number.

C. Extended aperture diagnostic (family EXT — a separate experiment)

Family EXT is a *different* experiment: $P = 30$ only, seven SNR points extended down to -10 dB, and different trial counts (Table II). Its numbers must not be substituted for the B3 numbers above, and we report both because they answer different questions — B3 gives the paired gain across two pilot counts, EXT gives the absolute NMSE of each estimator separately.

Fig. 3 plots the two absolute NMSE curves. Two facts separate cleanly. **EM-GS is flat in** N: SNR-averaged -10.321, -10.337, -10.301 dB at $N = 8, 16, 32$, a spread of 0.036 dB, and equally flat inside every panel. **HS-GS is the only curve that moves**: -10.350, -11.149, -12.753 dB, so the gap widens monotonically, $0.03 \to 0.81 \to 2.45$ dB.

These are not the B3 numbers and should not be quoted as such. The difference ($0.03/0.81/2.45$ versus $-0.19/+0.78/+2.85$) is fully explained by configuration: EXT averages seven SNR points including -10 dB at one pilot count, B3 averages twelve points at two pilot counts. Both are internally consistent; neither is more correct.

In the EXT configuration the $N = 8$ sign change is again visible: the SNR-averaged 0.03 dB conceals HS-GS gaining 0.744 dB at -10 dB but losing 0.190–0.403 dB at 0 dB and above.

X. Controlled Path-Count Experiment

Every experiment above draws L_k at random, so none isolates path count. Family EXT fixes $L_k = L$ identically across users and sweeps it to the capacity.

At $N = 32$, $P = 30$, SNR $= 5$ dB, RSR $= 12$ dB, 300 trials per point, the gain decays strictly monotonically:

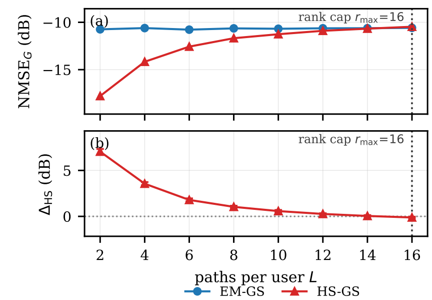

Fig. 4. **Family EXT.** Controlled path count: $N = 32$, $K = 3$, $P = 30$, SNR $= 5$ dB, RSR $= 12$ dB, 300 trials per point, L fixed and identical across users. (a) EM-GS is flat in L — path count does not affect an estimator that ignores structure — while HS-GS rises to meet it. (b) The paired gain decays monotonically to zero as L approaches $r_{\max} = \lceil N/2 \rceil = 16$; the 95% paired bootstrap band is plotted but is narrower than the markers. The hypothesis was recorded before the experiment was run.

$$7.04,\ 3.56,\ 1.79,\ 1.04,\ 0.58,\ 0.27,\ 0.05,\ -0.12 \text{ dB} \quad \text{for } L = 2, 4, \ldots, 16,$$

with the per-trial win rate falling from 100% to 45.0% and EM-GS flat to 0.191 dB across the entire sweep.

This is the mechanism test, and it rules out generic denoising. A generic denoiser would keep helping at large L, where the estimate is exactly as noisy as at small L. Instead the gain vanishes precisely where the prior becomes vacuous, $L \to r_{\max} = 16$. The flat EM-GS curve confirms the sweep is not confounded by the baseline changing.

Two details, both negative. The null is reached at $L = 14$ ($+0.05$ dB, CI $[-0.05, +0.14]$), one grid point *before* the ceiling, and at $L = 16$ the gain is small but significantly *negative* (-0.12 dB, CI $[-0.21, -0.04]$). The selector explains this rather than the theory: mean $\hat{L}$ reaches only 11.63 at $L = 16$, so the constraint is still active in 89.3% of trials and there discards genuine components — the $\hat{L} < L$ under-selection regime of §V-D.

XI. Structural Spectrum Diagnostic

Fig. 5 shows, for one illustrative realisation, why the prior exists and what the projection actually achieves.

The true channel's Hankel spectrum falls to $\sim 10^{-16}$ immediately after index L: the low-rank property is exact, as (7) states. The EM-GS estimate has no such cliff — noise fills the spectrum, so the estimate is not a sum of L exponentials and does not lie in $\mathcal{S}_L$. After four Cadzow sweeps a cliff reappears at the correct index, but only to $\sim 10^{-3}$.

That shortfall is the concrete measure of (10) being an approximate rather than exact projection, and it upper-bounds how much structure the method can impose — consistent with HS-GS attaining only part of the CCRB headroom (§VIII-B).

Caveat. This is one realisation chosen for illustration. It shows the qualitative mechanism; it is not a statistical claim about the magnitude of the residual, and we do not use it as one.

XII. From Path Count to Effective Rank

§X establishes the mechanism in terms of the generator parameter L. That parameter has two defects as a general

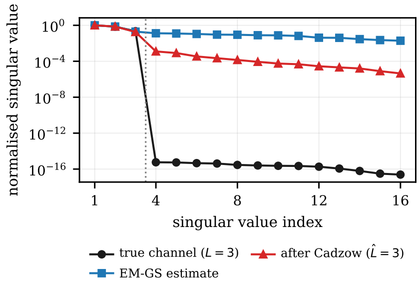

Fig. 5. **Family EXT, single illustrative realisation.** Normalised Hankel singular values for one channel column, $N = 32$, $L = 3$, SNR $= 5$ dB, RSR $= 12$ dB, $n_{\mathrm{cz}} = 4$, order forced to the true L. The true channel is exactly rank L ($\sigma_4/\sigma_1 = 5.7 \times 10^{-16}$), the EM-GS estimate is full numerical rank ($\sigma_4/\sigma_1 = 0.130$), and four Cadzow sweeps restore the cliff only approximately, to 1.2×10^{-3}.

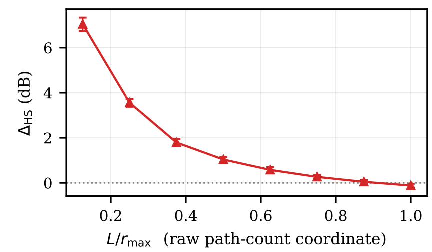

Fig. 6. **Family EXT.** The controlled path-count sweep of Fig. 4 plotted against the *raw* coordinate $L/r_{\max}$ ($N = 32$, $r_{\max} = 16$, $P = 30$, SNR $= 5$ dB, RSR $= 12$ dB, 300 trials per point; error bars 95% paired bootstrap). Compare Fig. 7, where the same physical effect is indexed by effective rank and becomes comparable across array sizes.

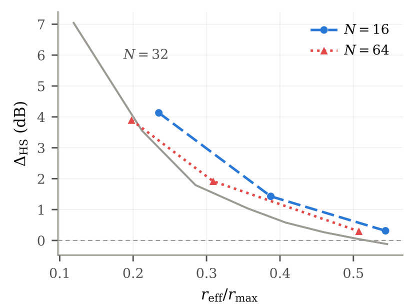

Fig. 7. **Family TD.** Δ_{HS} against $\rho = r_{\mathrm{eff}}/r_{\max}$ for $N \in \{16, 32, 64\}$. $K = 3$, $P = 20$, RSR $= 10$ dB, $T = 100$, $n_{\mathrm{cz}} = 1$, adaptive order, SNR $\sim \mathcal{U}[-10, 20]$ dB, paired per-trial median; 83–874 trials per cell. The grey curve is the re-indexed $N = 32$ relation. The sign boundary is stable to within 0.070 in ρ; the gain *magnitude* is not, with a one-signed residual of mean 0.493 dB (max. 0.901 dB) between $N = 16$ and $N = 64$.

coordinate.

First, L **is not a property of a channel** but of the process that produced it; a receiver cannot measure it, and two generators with the same L need not produce equally compressible liftings. Second, and decisively, L **mispredicts**: a 16-path channel on a 32-element array is algebraically full rank, so raw path count says the prior is vacuous, yet its effective Hankel rank is 8.73 and the lifting is in fact compressible. Closely spaced paths (§IV) produce exactly this situation.

We therefore replace L by a spectral measure. For singular values σ_i of $\mathcal{H}(\mathbf{g})$ define $p_i = \sigma_i / \sum_j \sigma_j$ and the Roy–Vetterli effective rank [7]

$$r_{\mathrm{eff}} = \exp\Big(-\sum_i p_i \log p_i\Big), \tag{20}$$

normalised by the capacity (8) to give the dimensionless coordinate

$$\rho = r_{\mathrm{eff}}/r_{\max}. \tag{21}$$

r_{eff} **must be computed on the noiseless channel.** The effective rank of the *estimate* is set by the noise floor, not by the channel: it was measured at 8.91–11.77 across $L = 1 \ldots 16$ at 5 dB, which would collapse every configuration onto essentially one point and destroy the coordinate. This is a diagnostic quantity computed from a known channel, not something a receiver estimates online.

Fig. 6 shows the path-count sweep on the raw $L/r_{\max}$ axis for comparison with the re-indexed version in Fig. 7. Re-indexing the same sweep by ρ moves the zero crossing from $L/r_{\max} = 0.90$ to $\rho = 0.518$, and turns a statement about one simulator's L values into one about any channel: the useful region is $\rho \lesssim 0.5$.

XIII. Array-Invariant Effective-Rank Boundary

If ρ is the right coordinate, curves measured at different array sizes should line up on it. Family TD tests this at $N \in \{16, 32, 64\}$ (Fig. 7).

The sign boundary is stable. The zero crossings are $\rho = 0.588$, 0.518, 0.544 at $N = 16, 32, 64$ — a spread of 0.070 across a fourfold change in aperture. Within the tested range, $\rho \approx 0.5$ locates where the prior stops paying, irrespective of array size.

The gain magnitude does not collapse, and we state the residual rather than report a collapse. Over the ρ window both arrays cover, $N = 16$ lies above $N = 64$ by a mean of 0.493 dB and a maximum of 0.901 dB, one-signed throughout and monotone in N. **We have no account of this residual.** One candidate is pencil quantisation at small capacity — $r_{\max}(16) = 8$ admits eight distinct ranks against 32 at $N = 64$, so the selector's grid is coarser — but that is untested, and the experiment that would settle it is a pencil sweep at fixed N.

Two of the three crossings are extrapolated. Neither $N = 16$ nor $N = 64$ has a measured cell whose gain is actually negative; their crossings are obtained by extending the last two measured points. Only the $N = 32$ crossing is bracketed by a measured sign change.

What we therefore claim. This is an *empirical effective-rank relation* over the tested array sizes and channel families. It is not a scaling law, it is not universal, and the imperfect magnitude collapse is a genuine limitation rather than a rounding detail.

XIV. Prospective Cross-Generator Validation

A. Chronology, and what makes this prospective

An earlier stage of this work concluded that robustness under clustered propagation was not established and named the decisive next experiment: *repeat the principal comparison under a clustered generator — a falsification test, not a confirmation exercise.* This section is that test, run twice.

The order of operations matters and was enforced by version control. For each new generator: (i) r_{eff} was measured on its noiseless channels and ρ computed; (ii) a predicted Δ_{HS} was read off the relation and **committed to the repository**; (iii) only then was the estimator run. **The relation was not refitted after any measurement.**

B. The prediction rule, stated so it can be reproduced

The relation is the eight-point table of $(\rho, \Delta_{\mathrm{HS}})$ pairs obtained by re-indexing the path-count sweep of §X. The predicted Δ_{HS} is its **piecewise-linear interpolation** at the measured ρ — `numpy.interp` on those eight points. **No curve is fitted**: there is no regression, no spline, no parametric form. The prediction interval is that same interpolation evaluated at the interquartile endpoints of r_{eff} over the channel realisations, so it reflects channel variability rather than being asserted.

C. Terminology: cross-generator, not out-of-model

Both validation generators produce receive-side ULA channels that are sums of spatial complex exponentials, i.e. they still satisfy (2). What differs is the *distribution* over $(L_k, \alpha_{\ell,k}, \theta_{\ell,k})$ — clustered rather than uniform angles, a different path count, different gain statistics. We therefore call these **cross-generator** or distribution-shift tests. Calling them “out-of-model” would overstate what is being tested: the mathematical family is unchanged.

This does not weaken the result. The relation was fitted using one channel-generation distribution and tested prospectively, without refitting, on previously unseen channel-generation distributions.

D. Results

Table III and Fig. 8 give the outcome.

Clustered Saleh–Valenzuela [13]. At $\rho = 0.331$ the relation predicts $+1.30$ dB. Measured: $+1.173$ dB, CI $[+1.017, +1.380]$ — an error of 0.13 dB, with the prediction inside the measured interval.

A second, independently specified configuration [14]: the receive-side Saleh–Valenzuela configuration of a precoding study by the same group as [1] — $L = 10$ paths, $\alpha_\ell \sim \mathcal{CN}(0,1)$, angles $\mathcal{U}(-90°, 90°)$. Only the channel configuration is borrowed; that work is not a competing estimator, and it uses ULAs at both ends with incident and departure angles, of which we use the receive side only. At $\rho = 0.400$ the relation predicts $+0.648$ dB with interval $[+0.367, +1.015]$. Measured: $+0.963$ dB, CI $[+0.808, +1.086]$.

The prediction error here is 0.315 **dB** — inside the registered interval and on the correct side of zero, but substantially larger than the first. We report it as such and not as a second 0.13 dB hit. Together the two bound what the relation is good for: it places an unseen generator on the correct part of the curve, and it is *not* accurate to a tenth of a decibel in general.

TABLE III
PROSPECTIVE PREDICTIONS AGAINST MEASUREMENTS. EACH PREDICTION WAS COMMITTED TO VERSION CONTROL BEFORE THE CORRESPONDING MEASUREMENT WAS RUN, AND THE RELATION WAS NOT REFITTED AFTERWARDS. FAMILY TD CONFIGURATION.

Generator	ρ	Pred.	Meas.	CI_{95}	Err.
Clustered SV [13]	0.331	$+1.30$	$+1.173$	$[1.017, 1.380]$	0.13
$L = 10$ cfg. [14]	0.400	$+0.648$	$+0.963$	$[0.808, 1.086]$	**0.315**
Literal reading	0.748	-0.12	0.000	degen.	0.12

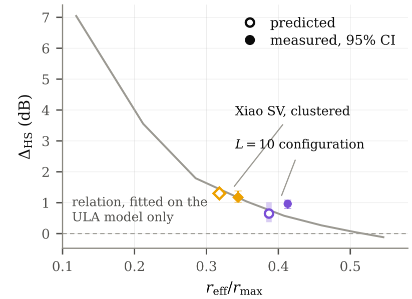

Fig. 8. **Family TD.** Predictions registered before measurement, against measurement. $N = 32$, $K = 3$, $P = 20$, RSR $= 10$ dB, adaptive order, $n_{\mathrm{cz}} = 1$, SNR $\sim \mathcal{U}[-10, 20]$ dB; 266–276 trials per generator. Both generators produce channels of the form (2); what differs is the distribution over $(L_k, \alpha_{\ell,k}, \theta_{\ell,k})$. The relation was fitted on the generator of §X alone and not refitted afterwards.

A third reading, and a partial match. Interpreting the clustered specification literally — all 40 rays drawn independently rather than in clusters — raises ρ to 0.748, where the relation predicts a small negative gain, -0.12 dB. The measured median is exactly 0.000 dB. The magnitude is right to 0.12 dB, but *the mechanism is not the predicted one*: rather than degrading, the order selector declines to project at all in 29.1% of trials, against 1.4% on the clustered model and 1.5% on the second configuration. The rule discriminates between generators without being told which it faces. We record this as a partial match.

Clustering did not eliminate the gain, which the earlier stage had identified as a live possibility.

XV. USER COUNT AND K-INVARIANCE

Equation (9) showed K cancelling from the compression ratio, motivating the hypothesis that Δ_{HS} is invariant in K. Testing it requires care: sweeping K at fixed P would simultaneously change *pilot adequacy* $P/2K$ and confound the two effects. We therefore hold $P/2K = 3.33$ fixed, using $P = 13, 20, 27$ at $K = 2, 3, 4$.

The outcome is genuinely mixed and we report it that way (Fig. 9).

At SNR ≥ 5 dB the invariance holds well: Δ_{HS} spans $+3.563$ to $+3.658$ dB across $K \in \{2, 3, 4\}$, a spread of 0.095 dB.

Pooled over all SNR it does not: the spread is 0.294 dB and is monotone in K ($+3.403$, $+3.161$, $+3.109$ dB). Our preregistered falsifier was a monotone trend exceeding 0.3 dB, **which this misses by** 0.006 **dB**. We do not present that as a

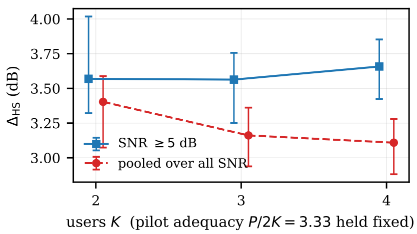

Fig. 9. **Family TD.** Δ_{HS} against user count at *fixed* pilot adequacy $P/2K = 3.33$ ($P = 13, 20, 27$ at $K = 2, 3, 4$). $N = 32$, RSR $= 10$ dB, $T = 100$, $n_{\text{cz}} = 1$, adaptive order, SNR $\sim \mathcal{U}[-10, 20]$ dB; 245–306 trials per cell; paired per-trial median with 95% bootstrap intervals. The high-SNR spread is 0.095 dB; pooled over all SNR it is 0.294 dB and monotone, missing the pre-registered 0.3 dB falsifier by 0.006 dB.

pass. The adjacent-pair confidence intervals overlap throughout, so no individual step is significant, and the pooled trend is driven by the bins below 0 dB where each cell contributes roughly 45 trials.

The honest statement is that the DoF cancellation transfers to estimation performance at moderate and high SNR, that below 0 dB there is a decline with K of about 0.3 dB that (9) does not predict, and that the low-SNR evidence is thin enough to warrant re-measurement before being believed.

XVI. Summary of Quantitative Findings

Table IV collects the principal numbers with their experiment family, so that no result is read against the wrong configuration.

XVII. Discussion

The experiments are not a list; they build on one another, and it is worth saying explicitly which statements are derived, which are observed, and which are interpretation.

Derived. The rank bound (7), the capacity (8), the compression ratio (9), the rank-one form of the Fisher information (15), and its block diagonality across receive elements. These follow from the model and require no measurement.

Observed. That HS-GS improves on EM-GS where the lifting is compressible; that the improvement grows with aperture while EM-GS does not move; that the gain decays to zero exactly as $L \to r_{\max}$; that the sign boundary sits near $\rho \approx 0.5$ across three array sizes; that two prospective predictions landed within 0.13 and 0.315 dB; that K-invariance holds at high SNR and not pooled.

Interpretation. The following is our reading, not a proof. The path-count experiment establishes that HS-GS benefits from Hankel compressibility rather than from generic denoising — the vanishing of the gain at the capacity is the discriminating observation. Effective rank then generalises that from a generator parameter to realised spectral complexity, which is what allows the relation to be carried to a generator it was not fitted on. The aperture result is explained by the two moving in opposite directions: $r_{\max} = \lceil N/2 \rceil$ grows with the array while the channel's effective complexity need not, so ρ falls and the prior becomes more informative. Finally, the Fisher analysis is what licenses attributing this to cross-element structure rather than to EM-GS degrading with N — and the measured 0.012 dB flatness is the empirical check on that attribution.

What would falsify the picture. If a cross-element prior with no Hankel structure produced the same aperture scaling, the mechanism claim would need revision. If the magnitude residual of §XIII turned out to scale with something other than pencil quantisation, ρ would be incomplete as a coordinate. If a generator with $\rho \ll 0.5$ produced no gain, the relation would be refuted outright; none of the three tested did.

XVIII. Limitations

Stated as a list of what has and has not been demonstrated.

1) **Simulation only.** No measured hardware, no recorded channel data.
2) **No hardware channel.** The atomic response is modelled through (1); non-idealities of a physical vapour cell are absent.
3) **Geometric ULA dependence.** All results assume (2); the cross-generator tests change the distribution but not the family.
4) $N = 8$ **fails at high SNR.** The gain is negative above 0 dB and the array-size result there is mixed, not weakly positive.
5) **Hard rank truncation is biased.** The bias does not vanish as noise decreases, which is exactly why $N = 8$ degrades at high SNR.
6) **One shared scalar rank for** K **users.** The selector applies a single $\hat{L}$ to all columns despite their differing true orders.
7) **Finite Cadzow sweeps.** Four sweeps leave a measurable residual (§XI); the structure is imposed only approximately.
8) **Non-convex structured projection.** The intersection of the rank and Hankel sets is a non-convex variety and (10) is not an exact idempotent projection onto it.
9) **No convergence guarantee** for the outer iteration (11), and we claim none.
10) **The effective-rank magnitude does not collapse across** N: a one-signed residual of mean 0.493 dB, maximum 0.901 dB, unexplained.
11) **Two of three zero crossings are extrapolated**, not bracketed by a measured sign change.
12) **The** K**-invariance pooled criterion narrowly fails**, missing the pre-registered falsifier by 0.006 dB.
13) **The second prospective prediction errs by** 0.315 **dB**, an order of magnitude worse than the first.
14) **The bounds do not govern the estimators compared to them.** CRLB and CCRB constrain unbiased estimators; every estimator here is biased.
15) **Cross-generator tests remain inside the ULA sum-of-exponentials family.** They are distribution shifts, not model violations.
16) **Equal user power throughout.** Near–far effects are untested.
17) **Limited array sizes and channel families**: $N \in \{8, 16, 32\}$ (B3, EXT) and $\{16, 32, 64\}$ (TD); three generators.

TABLE IV
PRINCIPAL QUANTITATIVE FINDINGS, EACH TAGGED WITH ITS EXPERIMENT FAMILY (TABLE II). NUMBERS FROM DIFFERENT FAMILIES ARE NOT COMPARABLE AS MEASUREMENTS OF THE SAME ESTIMATOR.

Finding	Value	Family	Section
Hankel capacity	$r_{\max}(N) = \lceil N/2 \rceil$	derivation	IV
Rank-two error in the bound	0.17–4.64 dB too low	B3 (36 pts)	VI
EM-GS vs unconstrained CRLB, SNR ≥ 5	within 0.051 dB	B3	VIII
EM-GS vs CRLB at -5 / 0 dB	0.295 / 0.237 dB	B3	VIII
CCRB below unconstrained CRLB	7.05–7.11 dB	B3	VIII
HS-GS over EM-GS, $N = 32$, $P = 30$	$+2.27$ to $+3.04$ dB	B3	VIII
Aperture gain, $N = 8/16/32$	-0.19 / $+0.78$ / $+2.85$ dB	B3 (12 pts)	IX
EM-GS flatness in N	0.012 dB	B3 (12 pts)	IX
Win rate, $N = 8/16/32$	33.5 / 74.0 / 95.3%	B3	IX
Constraint active, $N = 8/16/32$	57.5 / 92.7 / 99.6%	B3	IX
Aperture gap (separate experiment)	0.03 / 0.81 / 2.45 dB	**EXT**	IX
EM-GS flatness (separate experiment)	0.036 dB	**EXT**	IX
Path-count decay, $L = 2 \to 16$	$7.04 \to -0.12$ dB	EXT	X
EM-GS flatness in L	0.191 dB	EXT	X
Effective-rank zero crossings	0.588 / 0.518 / 0.544	TD	XIII
Magnitude residual, $N = 16$ vs 64	mean 0.493, max 0.901 dB	TD	XIII
Prospective error, generator 1 / 2	0.13 / **0.315** dB	TD	XIV
Abstention, clustered / literal	1.4% / 29.1%	TD	XIV
K-invariance, SNR ≥ 5 / pooled	0.095 / 0.294 dB	TD	XV
Runtime vs chained EM-GS	1.49 / 1.32 / 1.22×	B3	V-E
Order-selection share of runtime	57.7–85.5%	B3	V-E

18) **Order selection dominates runtime** (57.7–85.5%), and no attempt has been made to reduce it.
19) **The bias explanation of the CCRB crossings is circumstantial.** The retained `b3` archives store squared-error numerators only, so the per-trial shrinkage $c = \langle \hat{\mathbf{G}}, \mathbf{G} \rangle / \|\mathbf{G}\|_F^2$ that would make the argument quantitative cannot be recovered without rerunning the sweep (§VIII-B). An earlier draft quoted such coefficients; they are not reproducible from the stores retained here and are not restated.

XIX. CONCLUSION

The progression of this work is short to state.

Magnitude-only reception makes multi-user channel estimation a biased phase-retrieval problem. Under the unconstrained row-wise parameterisation of that problem the Fisher information is block diagonal across receive elements, so normalised per-element information does not grow with the array and an unstructured estimator gains nothing from aperture — which we then measure, at 0.012 dB of movement across a fourfold change in N. Exploiting aperture therefore requires cross-element structure, and the geometric ULA model supplies one: each path contributes a single spatial exponential, so the Hankel lifting of each column has rank at most the path count. Enforcing that rank inside the exact iteration, with the order chosen from held-out pilots, yields -0.19, $+0.78$, $+2.85$ dB at $N = 8, 16, 32$ and 2.27–3.04 dB at $N = 32$, $P = 30$ against a benchmark the baseline tracks to 0.051 dB.

A controlled path-count sweep establishes that this is structural rather than generic denoising, since the gain vanishes precisely where the prior becomes vacuous. Replacing raw path count by effective Hankel rank relative to capacity generalises the mechanism from a generator parameter to a measurable property of a channel, giving an empirical sign boundary near $\rho \approx 0.5$ that is stable to within 0.070 across $N \in \{16, 32, 64\}$ — though the gain magnitude does not collapse onto that axis, and we report the residual rather than the collapse. Used prospectively, with predictions committed before measurement and no refitting, that relation placed two previously unused channel generators on the correct part of the curve, to within 0.13 and 0.315 dB.

The claims are scoped accordingly. This is an empirical relation over the tested array sizes and channel families, not a law; the tested generators are distribution shifts within the ULA sum-of-exponentials family, not departures from it; the outer iteration has no convergence guarantee; the baseline is not efficient, only numerically close to a bound that does not strictly govern it; and the constrained bound is not a lower bound on the mean-squared error of a biased structured estimator. What has been demonstrated is that a measurable structural coordinate predicts, before the fact, when this prior is worth applying.

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